IGCSE 0607 Practice Questions — E9.2

Relative and expected frequencies

Total Questions: 15 — **Total Marks:** 104 Marks

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1 A biased five-sided spinner is numbered 1, 2, 3, 4, and 5.
The table shows the probability of the spinner landing on each number.

Number	1	2	3	4	5
Probability	0.15	0.25	x	0.2	$3x$

(a) Work out the value of x.

$x =$ ..... **[2]**

(b) The spinner is spun 250 times.
Calculate the expected number of times the spinner lands on a number greater than 3.

..... **[2]**

2 A spinner has five sections, labelled A, B, C, D and E.
The table shows the relative frequencies of the spinner landing on each section after 200 spins.

Section	A	B	C	D	E
Relative frequency	0.15	0.3	x	0.25	0.1

(a) Work out the value of x.

$x =$ ..... **[2]**

(b) Estimate the number of times the spinner lands on section B in 200 spins.

..... **[2]**

3 A biased 4-sided spinner, numbered 1, 2, 3, and 4, is spun 200 times.
The table shows the frequencies of each outcome.

Number	1	2	3	4
Frequency	45	35	70	50

(a) Write down the relative frequency of the spinner landing on 3.

..... **[1]**

(b) The spinner is spun 1000 times.
Estimate the number of times it will land on 1.

..... **[2]**

(c) Is the spinner fair?
Give a reason for your answer.

..... **[1]**

(d) Explain why spinning the spinner 1000 times gives a more reliable estimate of the probability than spinning it 200 times.

..... **[1]**

4 A biased 5-sided dice has faces numbered 1, 2, 3, 4, and 5.
Maya rolls the dice 200 times and records the frequency of each score in the table below.

Score on dice	1	2	3	4	5
Frequency	32	28	54	46	40

(a) Write down the relative frequency of rolling a score of 3, giving your answer as a decimal.

..... **[1]**

(b) Explain why the results in the table suggest that the dice is biased.

..... **[1]**

(c) Maya rolls the dice a further 500 times.
Calculate an estimate for the number of times she will roll an odd score.

..... **[3]**

(d) Maya rolls the dice n times and expects to roll a score of 5 exactly 36 times.
Find the value of n.

$n =$ ..... **[2]**

5 A biased four-sided die is rolled 60 times.
The scores are 1, 2, 3, and 4.
The frequencies of each score are shown in the table.

Score	1	2	3	4
Frequency	x	$2x + 5$	$3x - 2$	$x + 8$

(a) Find the value of x.

$x =$ ..... **[2]**

(b) Find the relative frequency of rolling a 3.

..... **[1]**

(c) The die is rolled 1200 times.
Calculate the expected number of times it rolls a 2.

..... **[2]**

6 A biased spinner has three sectors labelled A, B, and C.
In an experiment, the spinner is spun $(x^2 + 10x)$ times, where $x > 0$.
Sector A occurs $(2x + 4)$ times and sector B occurs $(5x + 2)$ times.
The relative frequency of the spinner landing on sector A is $\frac{1}{6}$.

(a) Show that $x^2 - 2x - 24 = 0$.

[2]

(b) Solve the equation $x^2 - 2x - 24 = 0$ and hence find the total number of times the spinner was spun.

Total spins = ... **[3]**

(c) Using the results of this experiment, find, as a fraction in its simplest form, the relative frequency of the spinner landing on

(i) sector B,

..... **[2]**

(ii) sector C.

..... **[2]**

(d) In a game, a player spins the spinner 600 times.
The player wins \$6 each time the spinner lands on sector A, wins \$2 each time it lands on sector B, and loses \$3 each time it lands on sector C. Calculate the player's expected net profit

after 600 spins.

\$ ..... **[3]**

7 A spinner has four colours: Red, Blue, Green, and Yellow.
A student spins it 50 times and records the results.
The table shows the relative frequencies for three of the colours.

Colour	Red	Blue	Green	Yellow
Relative frequency	0.15	0.45	0.10	x

(a) Find the value of x.

$x =$ ..... **[1]**

(b) The spinner is spun 300 times in total.
Estimate the number of times it lands on Blue.

..... **[2]**

(c) The spinner is meant to be fair.
Comment on whether the results show it is fair or biased.

..... **[1]**

8 A biased coin is tossed n times.
The probability of getting a head on each toss is p.
The expected number of heads is 30 and the expected number of tails is 45.

(a) Find the value of n.

$n =$ ..... **[1]**

(b) Calculate the value of p.

$p =$ ..... **[2]**

(c) The same coin is now tossed 300 times.
Work out the expected number of tails.

..... **[3]**

9 A bag contains a large number of coloured beads.
A bead is chosen at random, its colour recorded, and then returned to the bag.
This experiment is repeated 300 times.
The table shows the frequencies of the colours drawn.

Colour	Red	Blue	Green	Yellow
Frequency	84	105	x	y

(a) The relative frequency of choosing a Green bead is 0.22.
Work out the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[3]**

(b) There are exactly 15 Yellow beads in the bag.
Estimate the total number of beads in the bag.

..... **[2]**

(c) A new experiment is done where a bead is chosen and replaced 800 times.
Calculate the expected number of times a Blue or Green bead is chosen.

..... **[2]**

10 A researcher records the colours of cars passing a checkpoint.
Over one week, 800 cars pass the checkpoint.
The frequencies of the colours are shown in the table.

Colour	Silver	Black	White	Other
Frequency	260	180	w	$w + 40$

(a) Find the relative frequency of a car being Silver.
Give your answer as a fraction in its simplest form.

..... **[2]**

(b) Find the value of w.

$w =$ ..... **[2]**

(c) The researcher uses these results to estimate the probability that the next car passing the checkpoint is White.
Work out this estimate.

..... **[1]**

(d) In one month, 12000 cars are expected to pass the checkpoint.
Calculate the expected number of Black cars.

..... **[2]**

11 A quality control inspector tests a random sample of 250 light bulbs produced by a factory.
The table shows the lifetimes, t hours, of these 250 light bulbs.
The relative frequency of a light bulb having a lifetime of more than 1500 hours is 0.42.

Lifetime (t hours)	$0 < t \leq 500$	$500 < t \leq 1000$	$1000 < t \leq 1500$	$1500 < t \leq 2000$	$2000 < t \leq 2500$
Frequency	15	x	95	y	35

(a) Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[3]**

(b) A batch of 4800 light bulbs is produced by the factory.
Calculate an estimate for the number of light bulbs in this batch that have a lifetime of 1000 hours or less.

..... **[3]**

(c) Explain why it is important that the sample of 250 light bulbs is chosen at random.

..... **[1]**

(d) The inspector tests a second random sample of 750 light bulbs from the same factory, and finds that 115 of them have a lifetime of more than 2000 hours.
Combine the results of both samples to find the best estimate for the probability that a light bulb has a lifetime of more than 2000 hours.

..... **[2]**

12 A factory produces light bulbs.
The probability that a light bulb is defective is 0.025.
A sample of N light bulbs is taken.
The expected number of defective light bulbs in this sample is 40.

(a) Calculate the value of N.

$N =$ ..... **[2]**

(b) A new quality control process is introduced.
The factory takes a new sample of 2400 light bulbs.
The expected number of non-defective light bulbs in this new sample is 2364.
Calculate the percentage decrease in the probability of a light bulb being defective.

..... % **[5]**

13 A biased 5-sided spinner has sides numbered 1, 2, 3, 4, and 5.
The table shows the probability of the spinner landing on each number in terms of x.

Number	1	2	3	4	5
Probability	0.15	$2x$	$x + 0.05$	0.2	x

(a) Find the value of x.

$x =$ ..... **[2]**

(b) The spinner is spun 600 times.
Work out the expected number of times it lands on a prime number.

..... **[2]**

(c) A different spinner has 3 sections: Red, Blue, and Green.
The probability of landing on Red is p.
The probability of landing on Blue is $2p$.
The probability of landing on Green is $1 - 3p$.
This spinner is spun N times.

The expected number of times it lands on Red is 54 and the expected number of times it lands on Green is 36.
Work out the value of N and the value of p.

$N =$ ..... , $p =$ ..... **[4]**

14 A biased four-sided spinner is numbered 1, 2, 3 and 4.
The table shows the probability of scoring each number.
The spinner is spun 400 times.
Let E_1 be the expected number of times the spinner lands on 1.
Let E_3 be the expected number of times the spinner lands on 3.
You are given that $E_1 \times E_3 = 5600$.
Also, the probability of scoring a 3 is greater than the probability of scoring a 1.

Score	1	2	3	4
Probability	x	$2x$	y	y

(a) Show that $300x^2 - 100x + 7 = 0$.

[3]

(b) Solve the equation to find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(c) Calculate the expected number of times the spinner lands on an even number.

..... **[2]**

15 Two biased coins, Coin A and Coin B, are tossed.
The probability of Coin A landing on Heads is p.
The probability of Coin B landing on Heads is $p + 0.15$.
Coin A is tossed 300 times and Coin B is tossed 200 times.
The total expected number of Heads from these 500 tosses is 255.

(a) Form an equation in p and show that $p = 0.45$.

[3]

(b) Both coins are tossed together 1000 times.
Work out the expected number of times they land on different outcomes.

..... **[3]**

(c) In a new experiment, Coin A is tossed N times.
The relative frequency of landing on Heads is 0.48 and it lands on Heads 72 times.

(i) Work out the value of N.

$N =$ ..... **[2]**

(ii) Calculate the expected number of Tails in these N tosses, using the theoretical probability $p = 0.45$.

..... **[2]**